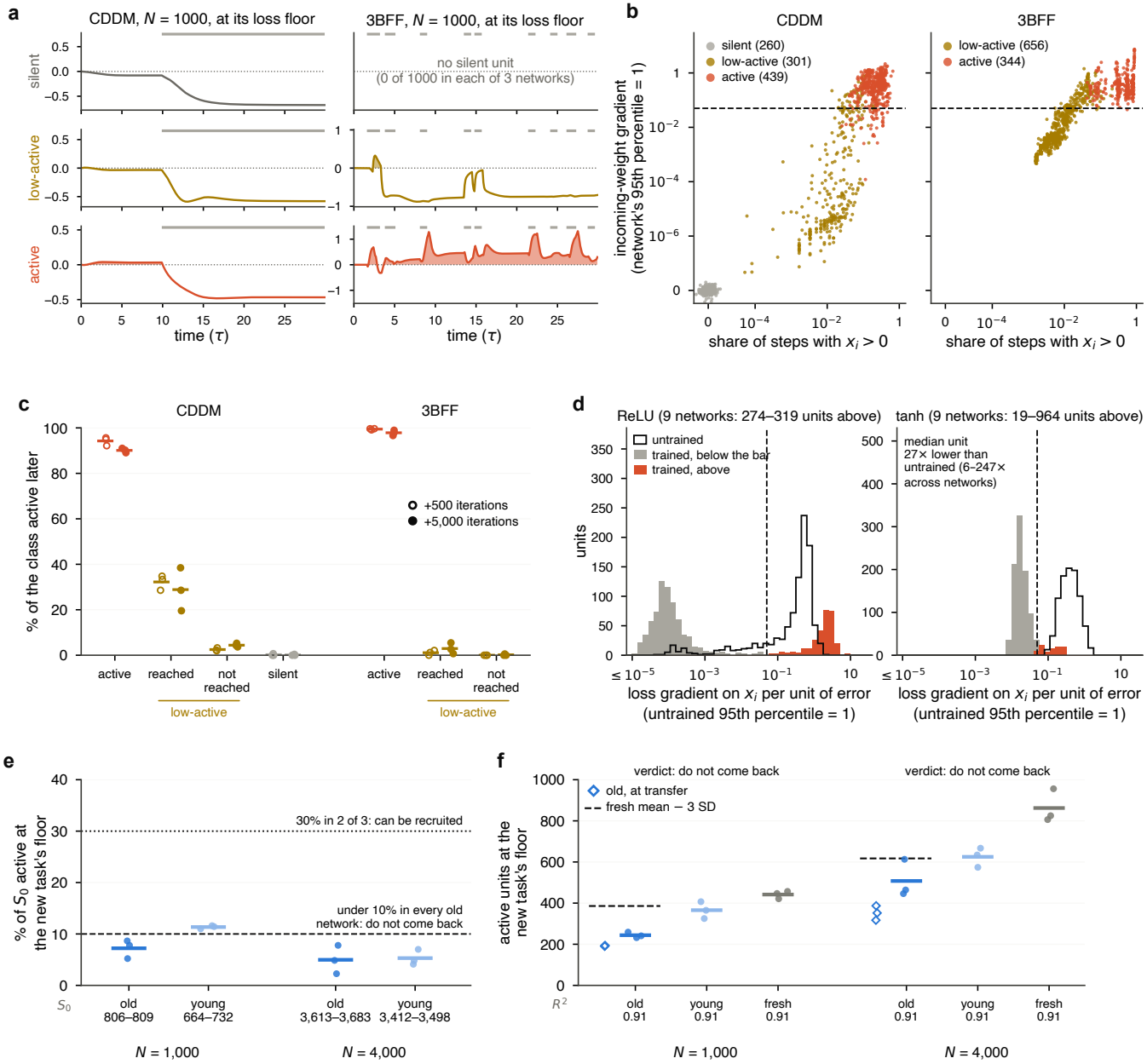

a
CDDM, N = 1000, at its loss floor
3BFF, N = 1000, at its loss floor
silent
low-active
active
no silent unit
(0 of 1000 in each of 3 networks)
time (τ)
b
CDDM
3BFF
silent (260)
low-active (301)
active (439)
low-active (656)
active (344)
incoming-weight gradient
(network's 95th percentile = 1)
share of steps with $x_i > 0$
c
CDDM
3BFF
% of the class active later
+500 iterations
+5,000 iterations
active
reached
not reached
silent
low-active
d
ReLU (9 networks: 274–319 units above)
tanh (9 networks: 19–964 units above)
untrained
trained, below the bar
trained, above
units
median unit
27× lower than
untrained (6–247×
across networks)
loss gradient on x_i per unit of error
(untrained 95th percentile = 1)
e
% of S_0 active at
the new task's floor
30% in 2 of 3: can be recruited
under 10% in every old
network: do not come back
S_0
old
806–809
young
664–732
old
3,613–3,683
young
3,412–3,498
N = 1,000
N = 4,000
f
verdict: do not come back
verdict: do not come back
old, at transfer
fresh mean − 3 SD
active units at the
new task's floor
R^2
old
0.91
young
0.91
fresh
0.91
old
0.91
young
0.91
fresh
0.91
N = 1,000
N = 4,000